NANYANG TECHNOLOGICAL UNIVERSITY
PH4501/PAP710 MIDTERM EXAMINATION

Tuesday, Oct 8th 2019

Total: 100 points. Time allowed: 60 minutes

1. **Ideal Gas (60 marks)**

   Consider the usual monoatomic ideal gas consisting of N particles.

   (a) Calculate the canonical partition function Z_N. Write your answer in terms of the thermal De Broglie wavelength $\lambda = h/\sqrt{2\pi m k_B T}$.

   (b) Find that the Helmholtz free energy F. Hence calculate the pressure P.

   (c) Calculate the grand canonical partition function $\Xi = \sum_N z^N Z_N$, where $z = \exp(\beta\mu)$ is the fugacity.

   (d) Calculate the average number of particles $\langle N \rangle$ in the grand canonical ensemble.

   (e) Using the grand canonical ensemble, show that the probability of finding the system to have N atoms is given by the Poisson distribution

   $$P_N = \frac{1}{N!} e^{-\langle N \rangle} \langle N \rangle^N$$

   (f) Using the canonical ensemble, show that

   $$\langle |p|^n \rangle = \frac{\int_0^\infty dp\, p^{n+2}\, e^{-\beta p^2/(2m)}}{\int_0^\infty dp\, p^2\, e^{-\beta p^2/(2m)}} = C_n (m k_B T)^{n/2}.$$

   Find the constant C_n in terms of the Gamma function $\Gamma(n)$. Find the value of C_0, C_1, and C_2 explicitly. Are your results sensible?

2. **Critical behavior of gas (40 marks)**

The equation of state of a special kind of gas is

$$P = Tn - \frac{b}{2}n^2 + \frac{c}{6}n^3,$$

where $n = N/V$ is the density, T is the temperature, and b and c are assumed to be positive temperature-independent constants.

(a) What is the second virial coefficient B_2?

(b) At the critical point, $\frac{\partial P}{\partial n} = 0$ and $\frac{\partial^2 P}{\partial n^2} = 0$. Solve these conditions to find the critical temperature T_c, the critical density n_c and the critical pressure P_c. Verify that $T_c n_c / P_c = 3$.

(c) Calculate the isothermal compressibility $\kappa_T = (1/n)\,(\partial n/\partial P)|_T$ as a function of T for $n = n_c$. Show that κ_T scales as $(T - T_c)^{-\gamma}$. What is the value of the susceptibility exponent γ?

(d) Define the shifted variables

$$\Delta P = \frac{P}{P_c} - 1, \quad \Delta T = \frac{T}{T_c} - 1, \quad \text{and} \quad \Delta n = \frac{n}{n_c} - 1.$$

Show that the non-dimensional equation of state is given by

$$\Delta P = 3(1 + \Delta n)\Delta T + (\Delta n)^3.$$

On the critical isotherm, $(P - P_c) \propto (n - n_c)^\delta$. In this case, what is the value of the critical exponent δ?

(e) The instability in the isotherms for $T < T_c$ is avoided by phase separation into a liquid of density n_+ and a gas of density n_-. For temperatures close to T_c, these densities behave as $n_\pm \approx n_c(1 \pm f(T))$. Using a Maxwell construction, or otherwise, find $f(T)$. Hence find the order parameter exponent β, which is defined by $(n_+ - n_-) \sim (T_c - T)^\beta$. Check that the critical exponents you calculated in this problem satisfy the Widom scaling law: $\gamma = \beta(\delta - 1)$.

Ideal Gas

(a) $$Z_N = \frac{Z_1^N}{N!}$$

$$Z_1 = \int \frac{d^3p\, d^3x}{h^3} e^{-\beta \frac{p^2}{2m}}$$

$$= V \int \frac{d^3p}{h^3} e^{-\beta \frac{p^2}{2m}}$$

$$= V \left[\int \frac{dp_1}{h} e^{-\beta \frac{p_1^2}{2m}} \right]^3$$

$$\sqrt{\frac{\pi}{\beta/2m}} \cdot \frac{1}{h}$$

$$= V \cdot \left(\frac{\sqrt{2m\pi k_B T}}{h} \right)^3 = \frac{V}{\lambda^3}$$

$$\Rightarrow \quad Z_N = \frac{V^N}{\lambda^{3N}} \frac{1}{N!}$$

b) $$F = -k_B T \log Z_N$$

$$= -k_B T \log \left[\frac{V^N}{\lambda^{3N}} \frac{1}{N!} \right]$$

$$p = -\frac{\partial F}{\partial V} = +k_B T \frac{\partial}{\partial V} \log V^N$$

$$= \frac{N k_B T}{V}$$

$$\Rightarrow \quad PV = N k_B T$$ ideal gas law

c) $$\Xi = \sum_{N=0}^{\infty} z^N Z_N \qquad z = e^{\beta\mu}$$

$$= \sum_{N=0}^{\infty} z^N \frac{V^N}{\lambda^{3N}} \frac{1}{N!} = \sum_{N=0}^{\infty} \left(\frac{zV}{\lambda^3}\right)^N \frac{1}{N!}$$

$$= \exp\left[\frac{zV}{\lambda^3}\right]$$

(d) $$\langle N \rangle = z \frac{\partial}{\partial z} \log \Xi = z \frac{\partial}{\partial z} \frac{zV}{\lambda^3}$$

$$= \frac{zV}{\lambda^3}$$

(e) Prob N particles $= P_N = \dfrac{z^N Z_N}{\Xi}$

$$= z^N \frac{V^N}{\lambda^{3N}} \frac{1}{N!} e^{-\frac{zV}{\lambda^3}}$$

$$= \frac{\langle N \rangle^N}{N!} e^{-\langle N \rangle}$$

which is the Poisson distribution.

(f) $\Gamma(z) = \int_0^\infty dy \; y^{z-1} e^{-y}$ Gamma function

$$\langle |p|^n \rangle = \frac{\int d^3p \; |p|^n e^{-\beta \frac{p^2}{2m}}}{\int d^3p \; e^{-\beta \frac{p^2}{2m}}}$$

$$= \frac{\int_0^\infty dp \; p^{n+2} e^{-\beta \frac{p^2}{2m}}}{\int_0^\infty dp \; p^2 e^{-\beta \frac{p^2}{2m}}}$$

let $y = \beta \frac{p^2}{2m} \Rightarrow p = \sqrt{\frac{2m}{\beta}} \sqrt{y}$

$dy = \frac{\beta}{2m} \; 2p\,dp = 2\sqrt{\frac{\beta}{2m}} \sqrt{y} \; dp$

$$\Rightarrow \langle |p|^n \rangle = \frac{\int_0^\infty \frac{1}{2}\sqrt{\frac{2m}{\beta}} \; dy \; y^{-1/2} \left(\sqrt{\frac{2m}{\beta}} \; y^{1/2}\right)^{n+2} e^{-y}}{\int_0^\infty dy \; \frac{1}{2}\sqrt{\frac{2m}{\beta}} \; y^{-1/2} \left(\frac{2m}{\beta} \; y^{1/2}\right)^2 e^{-y}}$$

$$= \left(\sqrt{\frac{2m}{\beta}}\right)^n \frac{\int_0^\infty dy \; y^{\frac{n+2}{2} - 1/2} e^{-y}}{\int_0^\infty dy \; y^{1/2} e^{-y}}$$

$$= (mk_BT)^{n/2} \cdot \underbrace{2^{n/2} \frac{\Gamma\left(\frac{n}{2} + \frac{3}{2}\right)}{\Gamma\left(\frac{3}{2}\right)}}_{C_n}$$

$$C_0 = \frac{\Gamma(\frac{3}{2})}{\Gamma(\frac{3}{2})} = 1$$

$$C_1 = 2^{1/2} \cdot \frac{\Gamma(2)}{\Gamma(\frac{3}{2})} = 2^{1/2} \cdot \frac{1}{\frac{1}{2}\sqrt{\pi}} = \sqrt{\frac{8}{\pi}}$$

$$C_2 = 2^{1} \cdot \frac{\Gamma(5/2)}{\Gamma(\frac{3}{2})} = 2 \cdot \frac{\frac{1}{2} \cdot \frac{3}{2} \cdot \sqrt{\pi}}{\frac{1}{2} \cdot \sqrt{\pi}} = 3$$

$$\frac{\langle |p|^2 \rangle_{\text{eq}}}{2m} = \frac{3mk_BT}{2m} = \frac{3}{2}k_BT$$

which is the classical equipartition theorem.

Critical behavior of gas

(a) $B_2 = -\frac{b}{2}$

(b) $\frac{\partial p}{\partial n} = T - bn + \frac{c}{2}n^2 = 0$

$\frac{\partial^2 p}{\partial n^2} = -b + cn = 0$

$\Rightarrow \quad n_c = \frac{b}{c}$

$\Rightarrow \quad T_c = b n_c - \frac{c}{2} n_c^2 = b \cdot \frac{b}{c} - \frac{c}{2}\left(\frac{b}{c}\right)^2$

$= \frac{b^2}{2c}$

$\Rightarrow \quad p_c = T_c n_c - \frac{b}{2} n_c^2 + \frac{c}{6} n_c^3$

$= \frac{b^2}{2c} \cdot \frac{b}{c} - \frac{b}{2}\left(\frac{b}{c}\right)^2 + \frac{c}{6}\left(\frac{b}{c}\right)^3$

$= \frac{b^3}{6c^2}$

$\Rightarrow \quad \frac{T_c n_c}{p_c} = \frac{\frac{b^2}{2c} \cdot \frac{b}{c}}{\frac{b^3}{6c^2}} = 3$

(c)

$$\kappa_P = \frac{1}{n} \left.\frac{\partial n}{\partial P}\right|_{T, n_c}$$

$$= \frac{1}{n} \cdot \left.\frac{1}{P - bn + \frac{c}{2}n^2}\right|_{n_c}$$

$$= \frac{1}{b/c} \cdot \frac{1}{P - b \cdot \frac{b}{c} + \frac{c}{2}\left(\frac{b}{c}\right)^2}$$

$$= \frac{c}{b} \frac{1}{P - \frac{b^2}{2c}} \qquad P_c = \frac{b^2}{2c}$$

$$= \frac{c}{b} \frac{1}{P - P_c} \sim (P - P_c)^{-1}$$

hence $\kappa_T \sim (P - P_c)^{-\gamma}$

$$\Rightarrow \boxed{\gamma = 1}$$

(d) $\underbrace{\Delta p = \frac{p}{p_c} - 1}_{p = p_c \Delta p + p_c}$ $\quad \underbrace{\Delta T = \frac{T}{T_c} - 1}_{T = T_c \Delta T + T_c}$ $\quad \underbrace{\Delta n = \frac{n}{n_c} - 1}_{n = n_c \Delta n + n_c}$

$$\Rightarrow \quad p_c \Delta p + p_c = (T_c \Delta T + T_c)(n_c \Delta n + n_c)$$
$$- \frac{b}{2}(n_c \Delta n + n_c)^2 + \frac{c}{6}\left(n_c(\Delta n + 1)\right)^3$$

$$\Rightarrow \quad \frac{b^3}{6c^2}(\Delta p + 1) = \frac{b^2}{2c}(\Delta T + 1)\frac{b}{c}(\Delta n + 1)$$
$$- \frac{b}{2}\left(\frac{b}{c}\right)^2(\Delta n + 1)^2 + \frac{c}{6}\left(\frac{b}{c}\right)^3(\Delta n + 1)^3$$

$$\Rightarrow \quad \Delta p + 1 = 3(\Delta T + 1)(\Delta n + 1) - 3(\Delta n^2 + 2\Delta n + 1)$$
$$+ (\Delta n^3 + 3\Delta n^2 + 3\Delta n + 1)$$

$$\Rightarrow \quad \Delta p + 1 = 3\Delta T \Delta n + 3\Delta T + 3\Delta n + 3$$
$$- 3\Delta n^2 - 6\Delta n - 3$$
$$+ \Delta n^3 + 3\Delta n^2 + 3\Delta n + 1$$

$$\Rightarrow \quad \boxed{\Delta p = 3(1 + \Delta n)\Delta T + (\Delta n)^3}$$

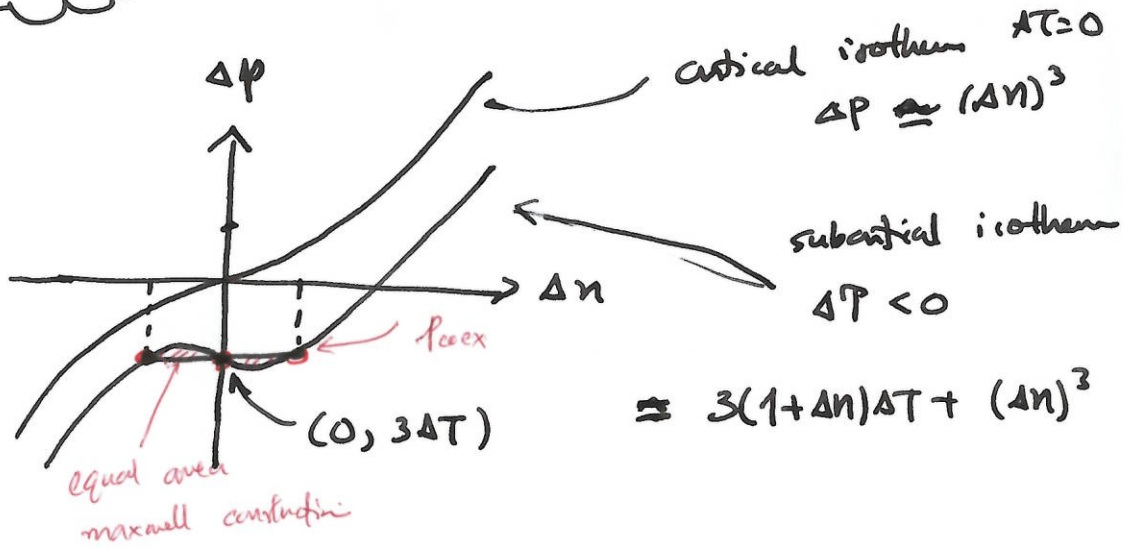
On critical isotherm $\Delta T = 0$

$$\Rightarrow \quad \Delta p = (\Delta n)^3$$

or $\quad p - p_c \propto (n - n_c)^3$

$$\Rightarrow \quad \boxed{\delta = 3}$$

(e)



$$\simeq 3(1+\Delta n)\Delta T + (\Delta n)^3$$

We see that $\Delta p = 3(1+\Delta n)\Delta T + (\Delta n)^3$ is "odd"

w.r.t. the point $(0, 3\Delta T)$

$$\Rightarrow \quad 3\Delta T = 3\Delta T + 3\Delta n\,\Delta T + (\Delta n)^3$$

$$\Rightarrow \quad (\Delta n)^3 + 3(\Delta n)(\Delta T) = 0$$

$$\Rightarrow \quad \Delta n = 0 \quad \text{or} \quad \pm\sqrt{-3(\Delta T)}$$

$$\Rightarrow \quad n_\pm = n_c \pm \sqrt{-3(\Delta T)}$$

$$\Rightarrow \quad n_+ - n_- \simeq 2\sqrt{-3(\Delta T)} \sim (T_c - T)^{1/2}$$

$$\Rightarrow \boxed{\beta = 1/2}$$

$\gamma = 1$

$\beta = 1/2$

$\delta = 3$

$$\Rightarrow \boxed{\beta(\delta - 1) = \tfrac{1}{2}\cdot 2 = 1 = \gamma}$$

Widom scaling law.